

Optimal Deviation from the Samuelson Rule

Pascal Michailat
<https://pascalmichailat.org/w/>

Optimal public expenditure.

$$1 = MRS_{gc} + m \times [1 - (-v'(\mu))]]$$

$$1 - MRS_{gc} = m \times [1 - (-v'(\mu))]]$$

$$\frac{1}{\epsilon} \times \frac{g/c - g/c^*}{g/c^*} = 2 \times \frac{u - u^*}{u^*}$$

Samuelson
spending
($1 = MRS_{gc}$)

elasticity of substitution
b/w
g & c

Formula in sufficient statistics.

$$\frac{1}{\epsilon} \times \frac{g/c - g/c^*}{g/c^*} = 2 \times m \times \frac{u - u^*}{u^*}$$

$\Rightarrow$

$$\frac{g/c - g/c^*}{g/c^*} = 2\epsilon \times m \times \frac{u - u^*}{u^*}$$

depends on g/c :
implicit
formula

c) Formula tells us how public expenditure g/c should deviate from benchmark given by Samuelson (1954) rule, g/c^*

Unemployment multiplier, m

Unemployment gap, $u - u^*$

$\backslash \quad \quad \backslash \quad \backslash$	$m < 0$	$m = 0$	$m > 0$ (more realistic)
$u - u^* < 0$ (too high) (boom)	$g/c > g/c^*$	$g/c = g/c^*$	$g/c < g/c^*$ (negative stimulus)
$u - u^* = 0$ (efficient)	$g/c = g/c^*$	$g/c = g/c^*$	$g/c = g/c^*$
$u - u^* > 0$ (too low) (slump)	$g/c < g/c^*$	$g/c = g/c^*$	$g/c > g/c^*$ (positive stimulus)

Δ Newer optimal to deviate from Samuelson enough so as to eliminate unemployment gap: only optimal to reduce $u - u^*$